Code-Animated Shorts from Generative AI Stills: A Practitioner Case Study on a DIY YouTube Channel

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Abstract

Solo creators increasingly combine generative AI with their own tooling to produce short-form video at volume. This practitioner report describes a pipeline built and run by one person: an AI image model produces still artwork, and all motion, captions and timing are written as deterministic HTML and JavaScript, rendered frame by frame in a headless browser and encoded with FFmpeg. We report the architecture, an engineering failure that matters to anyone rendering video from a browser (a memory leak in Playwright's synchronous API on Python 3.14), and an observational comparison on one YouTube channel. Eighteen Shorts produced with the pipeline between 15 and 20 October 2026 averaged 2.5K views (craft line, n = 7) and 1.2K views (illustrated-map line, n = 11), with 37% and 28% of viewers choosing to watch rather than swipe away. In the same month, 10–11-second craft Shorts built from AI-generated video clips reached 26K to 3.5M views with about 50% viewed. We discuss why reproducible, code-driven motion did not win in the short-form feed, where it remains useful, and the limits of single-channel evidence.

Keywords: generative AI; programmatic animation; short-form video; YouTube Shorts; creator economy; headless browser rendering

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1. Introduction

Generative video tools can produce rich, camera-like motion, but they are hard to control: timing cannot be matched exactly to narration, objects change between shots, and re-generating one shot changes its neighbours. Code-driven animation is the opposite: exact, reproducible and cheap to revise, but its motion is only as rich as the code.

We built a pipeline that uses generative models for still pictures and code for everything that moves, and used it to publish Shorts on an established craft and holiday-decor channel. This report asks a practical question: on a real channel, how did those Shorts perform against the channel's other short-form content?

Contributions:

1. A description of a working, single-operator pipeline (Section 3).
2. A documented failure and fix for frame-by-frame browser rendering on Python 3.14 (Section 4).

3. Observational audience data from 18 pipeline Shorts and the channel's comparison Shorts (Section 5).
4. Lessons on where code-driven motion fits, and on disclosure and platform policy (Section 6).

This is a case study, not a controlled experiment. The numbers describe one channel in one month.

2. Scope

This version is a practitioner report. It documents one system and one channel and does not attempt a literature review; a review of related work on generative AI in media production, short-form video audiences and AI-content disclosure is planned for version 2.

3. System description

3.1 Channel context

The channel (Do Huy Toan, @dhtoan) publishes wordless craft Shorts about glowing handmade decorations, plus some long-form videos on vintage holiday history. In the 28 days to 24 October 2026 it recorded 16.2 million views and about 21,000 new subscribers, reaching 36,945 subscribers. Its audience, as reported by YouTube Studio, was 85% female, 57% aged 55 or over, and 91.7% on mobile; the largest viewer countries were India, Brazil and Mexico, with the United States at 6.4%.

3.2 Two production lines

The pipeline produced two kinds of vertical Shorts:

- **Craft line** (20–28 s): four AI-generated stills per Short (materials, two making steps, finished piece glowing), animated with camera moves, step captions and a reveal, designed to loop.
- **Illustrated-map line** (35–45 s): a 90–110-word narrated script on a holiday-history fact, with a map whose camera, highlights, pins, counters and AI-generated chibi characters are keyed to the narration's word timings.

Stills came from ChatGPT image generation, with Canva's image generator as a fallback. Narration used Microsoft Edge neural text-to-speech. Facts for the map line were limited to claims verified in a sources file before scripting.

3.3 Motion as deterministic HTML

Each Short is an HTML page driven by a small specification file and a shared engine. The page exposes a seek function; the renderer opens it in headless Chromium through Playwright, sets each frame time, takes a viewport screenshot and pipes the frames to FFmpeg (libx264). Because each frame is a pure function of time:

- the same Short always renders to the same frames;
- a single caption or camera move can be fixed and re-rendered without regenerating any artwork;
- cues are taken from the narration's word timings, so motion lands on spoken words.

During rendering every network request outside 127.0.0.1 is blocked so that output cannot depend on remote assets.

3.4 Quality checks and human review

Before upload each Short is checked against a written list: stills at several cue points are inspected, overlapping characters are rejected, captions must stay inside safe areas, and safety rules apply to the craft line (for example, LED lights only, never candles inside paper or lace). Uploading and scheduling are done by hand after the owner approves the day's batch.

4. Engineering failure: a memory leak in Playwright's sync API on Python 3.14

Python 3.14 records, on every asyncio future, the set of tasks awaiting it. Under Playwright's greenlet-based synchronous API, finished tasks were never removed from a long-lived future, and each kept its result alive. With one screenshot per frame (about 6 MB as PNG at 1080p), a 1,740-frame render grew to about 10 GB, and three parallel renders froze a 16 GB machine. The asynchronous API is not affected.

Fix: a small module, installed once as a site-wide `.pth` hook, removes finished tasks from that future after each call; it does nothing on earlier Python versions or if Playwright's internals change. The renderer also lowers its own CPU priority, which Chromium and FFmpeg inherit on Windows, so that a 4-core machine stays usable during renders. On an idle machine render time is unchanged.

5. Results

All audience figures were read from YouTube Studio on 24 October 2026, or from the pipeline's daily snapshots of the same data.

5.1 Pipeline Shorts versus the channel's other Shorts

Group	n	Length	Mean views	Viewed (vs swiped away)	Subscribers gained	Share of views from Shorts feed
Pipeline, craft line	7	21–24 s	2.5K	36.7%	25 (total)	17–77%
Pipeline, map line	11	22–41 s	1.2K	28.2%	7 (total)	29–58%
Channel craft Shorts from AI video clips (October)	7 shown	8–11 s	26K – 3.52M (range)	~50%	+32 to +140 each	90–93%

The 18 pipeline Shorts together earned 30.8K views, less than one fifteenth of the channel's views on a single day at the time. The comparison group is not a random sample: it lists the channel's best-performing October Shorts, which were made a different way (AI-generated video clips assembled into 10-second loops). The comparison shows the gap between the pipeline and what the channel's audience rewarded, not a controlled effect size.

5.2 What differed

Three differences are confounded and cannot be separated with this data:

1. **Length.** Pipeline Shorts were two to four times longer. The best comparison Shorts were 10–11 s and were rewatched (average view duration 150–260% of length).
2. **Opening.** Pipeline Shorts opened with a text hook over a still image; map Shorts opened with an English-language history question. The comparison Shorts opened on materials or the finished piece in continuous camera-like motion, with no words.
3. **Topic fit.** Narrated English history did not suit an audience that mostly watches wordless crafts and is largely outside the United States.

6. Discussion

Where code-driven motion still fits. Exact sync to narration, maps and numbers, reproducible re-renders, and one timeline reused across languages. These matter more in long-form explainers than in a feed judged in the first second.

Where it did not. In a short-form feed, viewers of this channel rewarded rich, continuous motion and short, rewatchable loops. Animating still images with code did not reach that bar at our level of effort.

Disclosure and policy. Artwork in all the Shorts described here is AI-generated. YouTube asks creators to disclose AI-generated or altered content that looks realistic [3], and its monetisation policies exclude mass-produced, generic or repetitive content, renamed "inauthentic content" in July 2025 [4]; a pipeline built for volume has to vary topics and add real editorial work to stay within those rules. About one fifth of comments on the AI-video craft Shorts pointed out visual inconsistencies typical of generated video (patterns changing between shots, steps that do not match), which is a trust risk even when views are high.

Limitations. One channel, one month, no randomisation, a comparison group chosen by performance, and platform metrics the creator cannot audit. Results depend on YouTube's recommendation system at the time.

7. Conclusion

A single operator can build a reproducible pipeline that turns AI-generated stills into narrated, captioned Shorts. On a craft channel whose audience prefers short, wordless, motion-rich loops, those Shorts averaged one-tenth or less of the views of the channel's best Shorts made from AI video. We now use code-driven motion where precision matters and AI video where motion richness matters.

Acknowledgements and AI-use statement

Artwork in the videos described was generated with ChatGPT (OpenAI) and Canva; comparison Shorts used AI-generated video. The first draft of this paper was written with the help of an AI assistant (Claude, Anthropic) from the author's code, notes and YouTube Studio readings; the author checked every statement and is responsible for the content.

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